Bargavi Sivaraman

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Summary

Computer Science undergraduate with hands-on experience building production-ready software systems, data pipelines, and full-stack applications. Strong foundation in data structures, backend development, and system design. Experienced collaborating in research and engineering environments, with an interest in developing scalable software that impacts real-world systems.

Skills

Programming: Python, Java, SQL
Web & Backend: REST APIs, React, HTML, CSS
Software Engineering: Data Structures, OOP, Version Control, Testing, Debugging
Systems & Tools: Git, Logging, Error Handling, Data Validation
Data & ML Exposure: Pandas, Matplotlib, EDA
Collaboration & Work Style: Remote Collaboration, Task Tracking, Cross-Functional Teamwork

Professional Experience

ARCS Associate – Autonomy Research Center for STEAHM (NASA-Affiliated), CSUN - Northridge, CA — Aug 2025 – Present

- Designed and implemented modular Python data pipelines for sensor data processing.
- Developed validation frameworks to detect edge cases and improve system reliability.
- Refactored components to improve maintainability and scalability.
- Collaborated in an Agile-style engineering environment using Git-based version control.
- Improved data processing efficiency through automation and structured error handling.

Product & Systems Management – ETHIX (Early-Stage Startup) — June 2025 – Present

- Supported development and optimization of internal operational systems.
- Assisted in improving workflow efficiency through structured process tracking.
- Helped maintain and organize data used for inventory and order management.
- Collaborated with team members to translate business needs into structured system improvements.
- Contributed to documentation and process standardization efforts.

Front Desk & IT Assistant – Associated Students Children's Center, CSUN – Northridge, CA — Oct 2023 – Present

- Built and maintained structured datasets to support research and reporting workflows.
- Used Excel for data cleaning, validation, and analysis of large datasets.
- Assisted with documentation and organization of internal resources for better accessibility.
- Collaborated with cross-functional teams in a remote environment to support ongoing projects.
- Tracked tasks, deliverables, and timelines to support project coordination and process improvement.

Projects

Astro-Cultivator – NASA ARCS Research Project

- Built data processing pipelines to clean, validate, and transform sensor data.
- Performed data analysis to improve the accuracy and interpretability of results.
- Collaborated with researchers to translate requirements into working data workflows
- Implemented validation checks to identify inconsistent or noisy inputs before downstream analysis.
- Developed intermediate data representations to improve the interpretability of analytical outputs.
- Assisted in integrating backend data workflows with visualization or reporting components.
- Refactored pipeline components to improve readability, reusability, and maintainability.

Manufacturing Automation & Monitoring Platform (Full stack Project)

- Built and maintained a clean, responsive web interface using React, HTML, and CSS.
- Developed backend services using Python and REST APIs to support data-driven features.
- Designed backend components with modular architecture to support scalability and maintainability.
- Implemented logging, error handling, and testing to ensure maintainable code.
- Developed web-facing data workflows to surface analytical results in a clear and structured format.
- Ensured consistency between backend processing pipelines and user-facing outputs.

Education

California State University, Northridge (CSUN) – Bachelor of Science in Computer Science | Major GPA: 3.4 / 4.0 | **Expected Graduation: Dec 2026**

Relevant Coursework: Data Structures & Algorithms, Database Management, Software Engineering, Web Development, Artificial Intelligence, Machine Learning.